

Programming Assignment # 4, Knee Osteoarthritis (KOA)

Instructor: Dr. Ali Hassan

Due Date: 26 Nov 2025

Knee Osteoarthritis (KOA) Severity Prediction with X-ray Images

1 Learning Objectives

By completing this assignment, you will:

1. Explore and prepare a real-world medical imaging dataset for machine learning tasks.
2. Build and evaluate baseline models using both classical ML and deep learning approaches.
3. Apply appropriate evaluation metrics (beyond accuracy) suitable for imbalanced, ordinal data.
4. Implement simple explainability techniques (e.g., Grad-CAM, feature importance).
5. Write a clear and concise technical report summarizing your findings.

2 Dataset

The dataset for this assignment is the *Knee Osteoarthritis Severity Grading Dataset*, available on Mendeley Data: <https://data.mendeley.com/datasets/56rmx5bjcr/1>. It is derived from the Osteoarthritis Initiative (OAI) and contains knee X-ray images labeled with Kellgren–Lawrence (KL) severity grades (0–4).

3 Assignment Tasks

Part A: Data Exploration & Preparation

- Download and inspect the dataset.
- Report:
 - Number of images per class (KL grades 0–4).
 - Show at least 2–3 example images for each class.
 - Plot the class distribution (bar chart).
- Perform preprocessing: resize all images to a fixed size, normalize pixel values.

Deliverables: Plots and brief notes.

Part B: Classical ML Baseline (20%)

- Extract simple features from X-rays:
 - Option 1: Texture features (e.g., Local Binary Patterns, GLCM).
 - Option 2: Raw pixel intensities reduced with PCA.
- Train a **Random Forest** or **SVM** classifier.
- Evaluate using:
 - Accuracy,
 - Macro-F1,
 - Confusion matrix.

Deliverables: Metrics and plots with brief notes.

Part C: Deep Learning Baseline

- Use transfer learning with a pretrained CNN (e.g., ResNet18 or EfficientNet-B0).
- Input: cropped/resized images of size 224×224 .
- Apply basic data augmentation: horizontal flips, small rotations, brightness adjustment.
- Evaluate using:
 - Accuracy,
 - Macro-F1,
 - Cohen's κ (agreement with ordinal grading).
- Apply Grad-CAM to visualize regions influencing predictions.

Deliverables: Training logs, evaluation metrics, and Grad-CAM visualizations with brief notes.

Part D: Comparative Analysis

- Compare the classical ML baseline (Part B) with the deep learning baseline (Part C).
- Discuss:
 - Which approach performed better, and why?
 - Which KL grades were hardest to predict (based on confusion matrices)?
 - How class imbalance influenced results.
- Write a concise discussion (up to 2 pages).

Deliverables: File `analysis.pdf` or `analysis.md`.

Part E: Reproducibility & Reporting

- Submit clean, well-commented notebooks/scripts.
- Provide a 5-page technical report in PDF format, structured as follows:
 1. Introduction
 2. Dataset and Preprocessing
 3. Methods (Classical ML and Deep Learning)
 4. Results and Analysis
 5. Conclusion and Limitations

Optional (Bonus +10%)

For additional credit, attempt one of the following:

- Implement an **ordinal classification loss** (since KL grades are ordered).
- Combine classical ML and deep learning predictions through simple ensembling.

4 Submission Checklist

- `eda.ipynb`
- `classical_ml.ipynb`
- `deep_learning.ipynb`
- `analysis.pdf` or `analysis.md`
- Technical report (max 5 pages, PDF)

WARNING:

- Try not to copy code or report. Anybody caught doing so, will get zero marks.